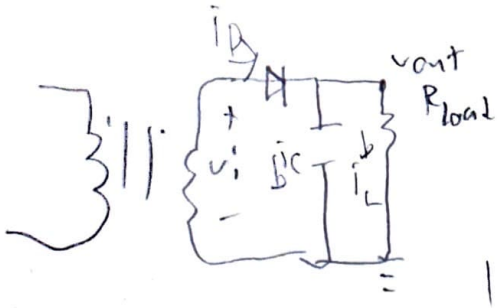
Dylan Thornburg Project 2 Prework

1.



The load current will depend on how much voltage gets stored in the cap.

when $v_i > .7$, $i_{load} = \frac{15.9\sin(120\pi t) - .7}{2500}$ A

$I_{peak} = \frac{15.9 - .7}{2500} = \boxed{6.08\,mA}$

$I_{peak} \approx I_L$ if the rectifier works correctly

$i_{Dmax} = i_L\left(1 + 2\pi\left(\sqrt{\frac{2V_p}{V_r}}\right)\right)$

31.8 p to p → 15.9 mag

120 v rms → 169.71 V mag

2. $\frac{120\sqrt{2}}{15.9} = 10.66 \sim 10$

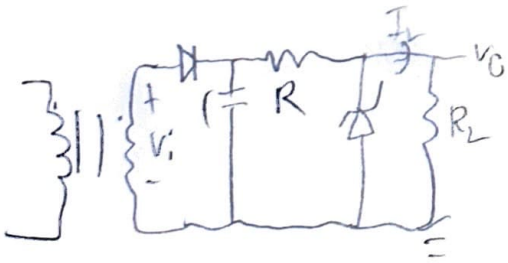
$\boxed{10.67:1 \text{ or } \sim 10:1}$

3. $V_{r(pp)} = \frac{1}{fRC}(V_p)$

$2 = \frac{1}{60(2500)C}(15.9 - .7)$, $C = 5.06 \times 10^{-5}$

$\boxed{C = 50.67\,\mu F}$

2.



dynamic zener res: 17Ω

test data point: 20mA

$5.1 = V_{20} + .02(17) \rightarrow V_{20} = 4.76$

$4.76 + \underset{(3.5mA)}{.0035}(17) = 4.8195$

15.9 - .7

$\frac{15.2 - 4.8195}{R} = .0035 + .00608$

$\boxed{R = 1083.56\,\Omega}$

Questions

1. I can use an oscilliscope to make sure my rectifier is cutting off negative (really <-.7V and slight ripple) voltages.

2. If the difference between the peak and lowest point of the wave form after the first period is less than the V_r specified, it is being met.

3. If I don't want to measure the load current, I could always measure load voltage and divide by resistance to check.

4. I would measure the V_r before and after adding the Zener diode and compare.

5. I would measure load regulation by using different resistors and seeing what happens to the voltage across the resistor. This will also tell me current after I take the measured voltage and divide it by R_{load}.

6. I would look up the datasheet for my load resistor or component and find the max load current for it. This will be my max load current for the whole psu